

```

    unsigned int rebase;
    struct module *mod;

    int apply_relocate_add Elf_Shdr *sechdrs,
        const char *strtab,
        unsigned int symindex,
        unsigned int relocs;
    struct module *mod;

    int module_finalize(const Elf_Shdr *hdr,
        const Elf_Shdr *sechdrs,
        struct module *mod);

    void module_arch_cleanup(struct module *mod);
}
#endif

```

- **rmmod** – remove a module from the kernel
- **depmod** – to find out the dependencies between modules
- **kerneld** – for invoking the *kerneld* daemon
- **kysms** – To display the symbols exported by the kernel, which the module can use
- **lsmod** – To list the loaded modules (see the code for the structure of its output)

```

aasiv@vayak@GNU-BQX-4 lsmod
Module                Size Used by
cbc                   3516 1052
aes_586               8124 1053
aes_generic           27484 1 aes_586
ecb                   2524 1
blkmisc              8356 1
ppdev                6688 0
vbomemadp            78760 0
vbomemdtl            85288 0
vbomdrv              121608 1 vbomemdtl
dm_crypt             12928 0
<<<CODE TRUNCATED>>>

```

- **modinfo** – To display the contents from the *modinfo* section. For example, the information about my video device can be displayed using the following code:
- ```

aasiv@vayak@GNU-BQX-4 modinfo videodev
filename: /lib/modules/2.6.31-17-generic-pae/kernel/drivers/media/
Video/Videodev.ko
alias: char-major-61.*
license: GPL
description: Device registrar for Video4Linux drivers v2
author: Alan Cox, Mauro Carvalho Chehab <mchehab@infradead.org>
reversion: 943A5776552DE0C369C89F77
depends: v4li-cunpbat
vermagic: 2.6.31-17-generic-pae SMP mod_unload modversions 586

```
- **modprobe** – helps you to load/remove modules in a better way. (We will discuss this later in this segment as well.) Here is the output that shows how *dummy.ko* is loaded into the kernel:

```

aasiv@vayak@GNU-BQX-4 strace modprobe dummy
execve("/sbin/modprobe", ["modprobe", "dummy"], 0 /* 36 vars */) = 0
brk(0) = 0x9d0c000
access("/etc/ld.so.nohwcap", F_OK) = -1 ENOENT (No such file or
directory)
mmap2(0, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_
ANONYMOUS, -1, 0) = 0xb770b000

```

```

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or
directory)
open("/etc/ld.so.cache", O_RDONLY) = 3
fstat64(3, {st_mode=S_IFREG|0644, st_size=89804, ...}) = 0
mmap2(NULL, 89804, PROT_READ|MAP_PRIVATE, 3, 0) = 0xb7680000
close(3) = 0
access("/etc/ld.so.nohwcap", F_OK) = -1 ENOENT (No such file or
directory)
open("/lib/libc.so.6", O_RDONLY) = 3
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0
200\1\0004\0
fsta64(3, {st_mode=S_IFREG|0755, st_size=1319364, ...}) = 0
mmap2(NULL, 1354416, PROT_READ|PROT_EXEC|MAP_PRIVATE|MAP_
DENYWRITE, 3, 0) = 0xb7582000
mmap2(0xb7680000, 12288, PROT_READ|PROT_WRITE, MAP_
PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x13e) = 0xb7680000
mmap2(0xb7680000, 10960, PROT_READ|PROT_WRITE, MAP_
PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xb7683000
close(3) = 0
mmap2(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_
ANONYMOUS, -1, 0) = 0xb75b1000
set_thread_area(&entry_number-1->6, base_addr:0xb75b18d0,
limit:1048575, seg_32bit:1, contents:0, read_exec_only:0, limit_in_pages:1,
seg_not_invent:0, useable:1}) = 0
mprotect(0xb7680000, 8192, PROT_READ) = 0
mprotect(0xb7683000, 4096, PROT_READ) = 0
mprotect(0xb772a000, 4096, PROT_READ) = 0
munmap(0xb7680000, 89804) = 0
uname(&sys="Linux", node="GNU-BQX-4") = 0
fstat64(2, {st_mode=S_IFCHR|0620, st_rdev=makedev(136, 0), ...}) = 0
brk(0) = 0x9d0c000
brk(0x9d0d000) = 0x9d0d000
open("/etc/modprobe.conf", O_RDONLY) = -1 ENOENT (No such file or
directory)
open("/etc/modprobe.d/_O_NONBLOCK|O_LARGEFILE|O_
DIRECTOR|O_CLOEXEC") = 3
fcntl64(3, F_GETFD) = 0x1 (flags FD_CLOEXEC)
getdents(3, /* 12 entries */, 32768) = 360
getdents(3, /* 0 entries */, 32768) = 0
close(3) = 0
open("/etc/modprobe.d/falsa-base.conf", O_RDONLY) = 3
fstat64(3, {st_mode=S_IFREG|0644, st_size=2497, ...}) = 0
mmap2(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_
ANONYMOUS, -1, 0) = 0xb770b000
read(3, "# autoloader aliases\ninstall so... 4096) = 2497
read(3, "", 4096) = 0
close(3) = 0
munmap(0xb770b000, 4096) = 0

```

I also suggest that beginners to take a look at the */proc/module* (please note, that this will get updated even if you leave it open):

```

mako@7830 ~$ cat /proc/module
fat 51452 1 mados, Live 0x5a5e000
cbc 3516 1163 - Live 0x5a32d000

```